Finite additive coding for an explicit Con(ZFC) equation

Ten unrestricted integer unknowns, degree 69 554 088 184,
and a secondary twelve-positive-variable equation of degree 938

Prepared for Vladimir Reshetnikov's ProveIt repository
Mathematical synthesis, explicit circuits, and reproducible audits

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Abstract

We give a fixed polynomial $F_{10} \in \mathbb{Z}[f, g, h, k, l, w, x, y, z, m]$ for which ZFC proves that the absence of integer zeros is equivalent to Con(ZFC), for the repository's calibrated proof predicate. The polynomial has exact total degree 69 554 088 184 and a division-free evaluator with 1 333 binary arithmetic operations. The existing ten-integer form in package 31 has degree $126496 \cdot 5^{31} + 27064$ and 1 442 operations. The degree therefore falls by a factor greater than $8.46 \cdot 10^{15}$, while the variable count is preserved and the evaluator becomes smaller.

The new mathematical step is to encode quartic monomials with a 32-element Bose–Chowla B_4 set instead of radix-five exponent positions. We prove the complete replacement lemma, including the off-center coefficient bound, arbitrary-input size bounds, and binary carry argument. Sun's ten-integer bridge and the repository's final relation combiner are retained with explicit attribution. A separate Pell-majorant improvement gives one equation in twelve positive integers of degree 938, or three equations of maximum degree 929.

The archive supplies all fixed coefficient payloads, executable arithmetic circuits, exact finite certificates, proof notes, and source provenance. These are ordinary mathematical reductions with explicit inherited dependencies; no new end-to-end Lean or Rocq certificate, proof of consistency, global minimality, or variable-count record is claimed.

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1 Result, scope, and comparison

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$. The statement Con(ZFC) means that there is no finite derivation of contradiction in the recursive ZFC proof system calibrated by ProveIt's K215 endpoint. The endpoint identifies a blank-input halt of the fixed machine with such a derivation. This convention fixes the arithmetized consistency statement; it is not an unspecified truth predicate or an existentially selected theory.

Theorem 1.1 (Main fixed instance)**.** *Let F_{10} be the polynomial defined in Sections 2–6, equivalently by the delivered file* `generated/equation10.json` *and its fixed bindings. Then*

$$\mathrm{ZFC} \vdash \Big[\exists (f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}^{10} \quad F_{10} = 0 \iff \neg\,\mathrm{Con(ZFC)}\Big].$$

Consequently ZFC proves

$$\boxed{\Big(\forall (f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}^{10},\ F_{10} \neq 0\Big) \iff \mathrm{Con(ZFC)}.}$$

The total degree is exactly 69 554 088 184*. Its delivered arithmetic circuit uses* 484 *additions,* 25 *subtractions, and* 824 *multiplications:* 1 333 *operations in total.*

All letters introduced as abbreviations denote explicit polynomials in these ten supplied variables. Circuit registers are not additional existential unknowns. There are no variable exponents, digit-extraction operations, divisions, radicals, unspecified indices, or free coefficient parameters in the equation.

The archive also gives a second result, proved in Section 8: twelve *strictly positive* unknowns, either a degree-938 equation with 122 operations or a three-equation system of maximum residual degree 929, with 117 operations. Positive and unrestricted integer domains must not be silently interchanged.

1.1 The reviewed baseline

The review is pinned to ProveIt commit

`22e75dd6b87b1b199bb08c778d7cf075ae5aae08`.

The relevant directory is

`SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms`.

The comparison includes the source-31 through source-37 refinements available at that commit, not merely an earlier universal equation. The decisive comparisons are as follows.

Presentation	Domain	Total degree / maximum degree	Operations
Source 31, one equation	10 in $\mathbb{Z}$	$126496 \cdot 5^{31} + 27064$	1 442
New additive code, same finalizer	10 in $\mathbb{Z}$	69 554 088 184	1 333
New code, literal Sun finalizer	10 in $\mathbb{Z}$	1 323 756 220 880	1 383
Source 34, low-degree equation	12 positive	1 022	124
New majorant, one equation	12 positive	938	122
Source 34, associated system	12 positive	1 013	120
New majorant, three equations	12 positive	929	117

The old ten-integer degree is exactly

$$589\,042\,901\,992\,797\,851\,562\,527\,064.$$

Its ratio to the new degree is approximately $8.4688465822 \cdot 10^{15}$. The evaluator saves 109 operations, about 7.56%. These improvements hold simultaneously at the same ten-integer variable count.

Other existing forms retain advantages in other metrics. In particular, the degree-1081, twelve-positive-variable evaluator takes only 108 operations; the new degree-938 presentation does not dominate that evaluator. Existing nine-positive-variable and low-degree many-variable forms remain relevant. The source-31 parent itself is a system of 22 quadratic equations in 31 positive variables. There is no single ordering called *simplest* that makes variable count, degree, circuit size, coefficient storage, and proof complexity equivalent. The new ten-integer result is a strict improvement on the reviewed ten-integer baseline in both reported complexity metrics.

1.2 What is new and what is inherited

The general reduction to ten unrestricted integer variables belongs to Sun [3]. The fixed consistency machine, its compiler, the source quartic, and the structural three-radical finalizer belong to the existing ProveIt construction [5]. Bose–Chowla additive sets are classical [1]. The contribution here is their application to the *entire bounded-degree monomial simplex*, together with a complete proof that the altered source code meets every downstream hypothesis. The source reversal target is also tightened to $4W$, and the fixed powers are shared explicitly.

The positive-variable improvement is independent of the additive code. It combines the complete coordinate shift in source 37 with the translated carrier in source 34, and proves a substantially smaller separating majorant. It uses the same complete coefficient tuple.

The claimed novelty is relative to the reviewed repository constructions. We do not infer worldwide priority from an absence of search results or claim an optimal B_4 set.

2 A completely fixed quartic source

Use the complete source-31 coefficient tuple associated with the 121×8801, 943-tile compiler. Its four equation ports are

$$\beta = \texttt{beta}, \quad K_0 = \texttt{K}, \quad M_C = \texttt{MC}, \quad M_F = \texttt{MF_source}.$$

The last port is the *already shifted* mask. Replacing only one member by a coefficient from a smaller tuple would invalidate the binding, so no such mixture is made here.

The payloads are nonnegative integers encoded as unsigned little-endian bytes and then gzip-compressed. Their exact bit lengths are

$$(1\,064\,921,\ 1\,070\,004,\ 1\,064\,921,\ 1\,064\,922).$$

They occupy 533 099 uncompressed bytes in total. The included coefficient builder reconstructs them from the supplied 943 Wang tiles and the 430 transitions of the fixed 215-state machine. The payloads, coefficient recipes, and manifests fix actual integers, not parameters to be chosen after deciding consistency.

Let $r_1, \dots, r_{22}$ be the quadratic residuals listed in Appendix A, with these four fixed ports. In the ordered 31 source coordinates define

$$P(n_1, \dots, n_{31}) = \sum_{j=1}^{22} r_j (n_1 + 1, \dots, n_{31} + 1)^2. \tag{1}$$

The shift is essential: the parent variables are positive, while the encoded variables are in $\mathbb{N}$. The inherited compiler and calibrated machine endpoint prove

$$\exists n \in \mathbb{N}^{31}\ P(n) = 0 \quad \longleftrightarrow \quad \text{blank K215 halts} \quad \longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}). \tag{2}$$

The included `KEndpoint215.lean` records the final theorem as `blank_halts_iff_zfc_refutation`; its imports and the full compiler are inherited dependencies, not newly rebuilt formal proofs.

The source polynomial has exact degree four. For example, the τ^4 coefficient in the sum of squares is nonzero. Also $P(0) > 0$: at the all-one parent assignment the fifth residual $E - X(a - E)$ is one. This positivity will make the reversed encoding polynomial positive even at inputs that have not yet passed the digit test.

Write

$$P(n) = \sum_{|\alpha| \le 4} p_\alpha n^\alpha, \qquad c_\alpha = (4 - |\alpha|)! \prod_{i=1}^{31} \alpha_i!\, p_\alpha.$$

The exact reconstruction produces 402 nonzero weighted records. Their coefficients are small polynomial recipes in the four fixed ports; after expanding those recipes there are 675 symbolic terms. Let $L = \max_\alpha |p_\alpha|$. The inherited, replayed coefficient bound is

$$|c_\alpha| + 1 < 2^{4\,269\,855}.$$

Retain the fixed extra constant

$$\boxed{\beta_S = 2^{4\,269\,882}}. \tag{3}$$

Since $48 \cdot 33^4 = 56\,924\,208 < 2^{27}$, it satisfies

$$\beta_S > 48 \cdot 33^4 L. \tag{4}$$

Its exponent is even. Both this strict coefficient margin and the congruence $\beta_S \equiv 1 \pmod 3$ will be used.

The authoritative new binding is `data/INSTANCE.json`. The upstream instance file is retained for fixed coefficients and provenance; its old radix-five exponents and old output paths are superseded.

3 Replacing positional exponents by a finite additive code

3.1 The exact property needed

A B_4 set is a set whose sums of four elements, with repetitions allowed and order ignored, uniquely determine their multisets. We need 32 elements, not 31: one is the zero padding coordinate for monomials of degree below four.

Fix the following ordered set:

$$\begin{aligned}(w_0, \dots, w_{31}) = (&0, 51609, 55430, 68596, 78030, 89463, 90233, 110498,\\ &110508, 128615, 136498, 211819, 241519, 305379, 312706, 314312,\\ &369115, 417382, 429683, 435112, 450230, 462314, 469608, 490199,\\ &501539, 507151, 524944, 550792, 559943, 573626, 607799, 610895).\end{aligned} \tag{5}$$

Thus $W = w_{31} = 610\,895$, and $w_0 = 0$ is not an unknown. The delivered checker verifies all $\binom{35}{4} = 52\,360$ unordered four-element multisets and confirms that their sums are distinct modulo $1\,048\,575$. This stronger modular property implies uniqueness of their integer sums.

Lemma 3.1 (Full-simplex injectivity)**.** *For $|\alpha| \le 4$, let*

$$e(\alpha) = \sum_{i=1}^{31} \alpha_i w_i.$$

Then e is injective on all such exponent vectors.

Proof. Represent α by α_i copies of w_i, padded by $4 - |\alpha|$ copies of zero. Equal sums imply equal four-element multisets. In particular, each positive weight has the same multiplicity, so the exponent vectors agree. □

Checking only collisions among the 402 source monomials would be insufficient. The other factor in coefficient extraction contains *every* monomial of total degree at most four. The lemma covers the whole simplex and therefore prevents those missing-source terms from contaminating the distinguished coefficient.

3.2 Origin and independent verification of the weights

The weights come from the classical Bose–Chowla construction [1]. Work in

$$\mathbb{F}_2[t]/(t^{20} + t^3 + 1),$$

and let θ be the residue of t. The supplied program checks that its successive powers enumerate all $2^{20} - 1$ nonzero bit patterns and return to one. This finite check shows at once that the quotient is a field and θ is primitive; no external table of primitive polynomials is assumed.

Put $Q = 2^{20} - 1 = 1\,048\,575$. The subfield of order 32 is

$$\{0\} \ \cup \ \{\theta^{33825j} : 0 \le j < 31\}.$$

For a in this subfield define $e_a \in [0, Q)$ by $\theta^{e_a} = \theta + a$. If two sums of four such logarithms agree modulo Q, the corresponding products of four factors $\theta + a$ agree. Their monic degree-four polynomials have a difference of degree at most three that vanishes at θ. Since θ has degree four over $\mathbb{F}_{32}$, that difference is zero. Unique factorization gives equality of the multisets.

Multiply all logarithms by the unit 208418 modulo Q, translate by -97920, and take the sorted representatives in $[0, Q)$. The result is (5). The translation removes a cyclic gap of length 437680. This was a finite search for a convenient set, not a proof that 610895 is the smallest possible diameter. The separate exhaustive multiset check certifies the literal list independently of the finite-field implementation.

3.3 The coefficient-isolation identity

Define the ordinary univariate polynomial

$$D_c(T) = \sum_{|\alpha| \le 4} c_\alpha T^{4W - e(\alpha)}. \tag{6}$$

Every exponent is nonnegative. Equality in $e(\alpha) \le 4W$ causes no problem, so a reversal target $4W + 1$, or the old $5W$, is unnecessary.

Lemma 3.2 (Quartic isolation and coefficient bound)**.** *For arbitrary $n_1, \dots, n_{31}$,*

$$[T^{4W}] \left(1 + \sum_{i=1}^{31} n_i T^{w_i}\right)^4 D_c(T) = 24P(n).$$

If all $n_i \geq 0$, every coefficient of this product has absolute value at most

$$24L \left(1 + \sum_i n_i\right)^4.$$

Proof. The term indexed by γ in the multinomial factor can contribute to degree $4W$ with the term indexed by α in D_c only if $e(\gamma) = e(\alpha)$. Lemma 3.1 gives $\gamma = \alpha$. Its multinomial coefficient is

$$\frac{24}{(4 - |\alpha|)! \prod_i \alpha_i!},$$

which cancels the factorial in c_α, proving the identity.

Different α's give different exponents of D_c, and $|c_\alpha| \leq 24L$. For any coefficient of the product, bound each contributing coefficient of D_c by $24L$, and sum all the nonnegative coefficients of the multinomial factor. Their sum is $(1 + \sum_i n_i)^4$. No factor counting the 402 source terms is needed. □

Because $P(0) > 0$, the leading coefficient of D_c is $24P(0) > 0$. For every integer B with $B - 1 > 24L$,

$$\left|D_c(B) - 24P(0)B^{4W}\right| \leq 24L \sum_{j=0}^{4W-1} B^j < B^{4W}.$$

In particular $D_c(B) > 0$. This estimate holds before any decoding assumption on the integer being tested.

4 The new two-variable source code

The only supplied variables in this section are $f, g \in \mathbb{Z}$. Put

$$\begin{aligned}
&b = 1 + 3f, && \mathcal{B} = \beta_S b^4, && \ell_c = \sum_{i=1}^{31} \mathcal{B}^{w_i},\\
&N_0 = \mathcal{B}^{W+1}, && N_1 = 4\mathcal{B}^{8W+1}, && N = N_0 N_1 = 4\mathcal{B}^{9W+2},\\
&M_c = N_0 - 1 - (b-1)\ell_c, && T_c = M_c + (\mathcal{B} - 2)\mathcal{B}^{4W} N_0, &&\\
&J_c = 2(1+g)^4 D_c(\mathcal{B}) + \mathcal{B} \sum_{j=0}^{8W} \mathcal{B}^j, && S_c = g + J_c N_0, &&\\
&R_c = (S_c + T_c + 1)N + T_c + 1, && X = (N-1)R_c, && Y = N^2,\\
&\mathscr{C} = b\mathcal{B}^W. &&&&
\end{aligned} \tag{7}$$

Here and below fixed-index sums and fixed integer powers are ordinary polynomial notation. In the circuit they are evaluated by binary additions and multiplications, without division.

Proposition 4.1 (Replacement source lemma)**.** *The formulas (7) have these properties.*

1. *If P has a nonnegative integer zero, one can choose f arbitrarily large, with b both a square and a power of two, and choose $b \le g < \mathscr{C}$, so that*
$$Y \mid \binom{2X}{X}.$$
2. *Conversely, $f \neq 0$, $0 \le g < 2\mathscr{C}$, b both a square and a power of two, and this divisibility imply that P has a nonnegative integer zero.*
3. *Even before assuming that b is a power of two, the conditions $f \neq 0$, b a square, and $0 \le g < 2\mathscr{C}$ imply*
$$0 \le S_c, T_c < N, \qquad 3 \mid X, \qquad X > 3b, \qquad Y > \max(b, 256).$$

4.1 Bounds before decoding

Since $b = 1 + 3f$ is a square and $f \neq 0$, it is at least four and $f > 0$. Inequality (4) gives $\mathcal{B} > 2b$. The base-$\mathcal{B}$ digits of M_c are $\mathcal{B} - b$ at the selected positions w_i, and $\mathcal{B} - 1$ at all other positions below $W + 1$. Consequently
$$0 \le M_c < N_0, \quad 0 \le g < N_0, \quad 0 \le T_c < N.$$
The bound on g uses $g < 2b\mathcal{B}^W < N_0$.

For arbitrary such g, not necessarily a correctly packed witness, $g + 1 \le 2b\mathcal{B}^W$. Let $A_c = 48L(2b)^4$. The coefficient margin gives $A_c < \mathcal{B} - 2$. Positivity and the geometric upper bound on D_c imply
$$\begin{aligned} 0 < J_c &\le A_c\mathcal{B}^{4W}\sum_{j=0}^{4W}\mathcal{B}^j + \mathcal{B}\sum_{j=0}^{8W}\mathcal{B}^j \\ &< \frac{(\mathcal{B}-2)+\mathcal{B}}{\mathcal{B}-1}\mathcal{B}^{8W+1} = 2\mathcal{B}^{8W+1} < N_1. \end{aligned}$$
Thus $0 \le S_c < N$ as well. Because $\beta_S \equiv b \equiv 1 \pmod 3$, we have $\mathcal{B} \equiv N \equiv 1 \pmod 3$ and hence $3 \mid X$. Finally $R_c \ge N + 1 > \mathcal{B} \ge b$, and the displayed sizes of N, X, Y give the strict inequalities in the proposition.

The cap retains exactly the form required by the downstream theorem:
$$\mathscr{C} = 2^{4\,269\,882W} b^{4W+1}. \tag{8}$$
Both exponents are fixed positive integers, and the exponent of two is even.

4.2 The digit mask and the distinguished coefficient

Assume now that b is a power of two. Then $\mathcal{B}, N_0, N$ are powers of two. Bitwise conjunction is denoted by & in the proof only. The literal digits of M_c imply
$$g \,\&\, M_c = 0 \quad\Longleftrightarrow\quad g = \sum_{i=1}^{31} n_i\mathcal{B}^{w_i} \quad \text{with } 0 \le n_i < b.$$
At selected positions the mask allows precisely the low $\log_2 b$ bits; elsewhere it allows none. This elementary mask statement is valid for arbitrary increasing positions, as in Sun's Lemma 2.4 [2]; see also [4].

For such decoded digits put

$$K_c = \left(1 + \sum_i n_i \mathcal{B}^{w_i}\right)^4 D_c(\mathcal{B}) + \frac{\mathcal{B}}{2} \sum_{j=0}^{8W} \mathcal{B}^j.$$

By (4) and Lemma 3.2, every coefficient before padding has absolute value strictly less than $\mathcal{B}/2$. Therefore each displayed base-$\mathcal{B}$ digit of K_c is an integer strictly between zero and $\mathcal{B}$, with no carries or borrows. Its digit at position $4W$ is $\mathcal{B}/2 + 24P(n)$. Among the integers strictly between zero and $\mathcal{B}$, only $\mathcal{B}/2$ has zero conjunction with $\mathcal{B}/2 - 1$. Thus

$$K_c \,\&\, ((\mathcal{B}/2 - 1)\mathcal{B}^{4W}) = 0 \quad \Longleftrightarrow \quad P(n) = 0.$$

Since $J_c = 2K_c$, shifting both operands by one binary bit gives

$$J_c \,\&\, ((\mathcal{B} - 2)\mathcal{B}^{4W}) = 0 \quad \Longleftrightarrow \quad P(n) = 0. \tag{9}$$

It would be incorrect to claim that doubling preserves all base-$\mathcal{B}$ digits without carries. Identity (9) uses a *binary shift*, so those possible base-$\mathcal{B}$ carries do not matter.

Because $g, M_c < N_0$, the low and high binary blocks separate:

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \,\&\, M_c = 0 \text{ and } J_c \,\&\, ((\mathcal{B} - 2)\mathcal{B}^{4W}) = 0.$$

The carry/binomial identity of Sun [2, Lemma 2.2], for any power of two N and $0 \le S_c, T_c < N$, states that the definition of R_c, X, Y in (7) gives

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad N^2 \mid \binom{2X}{X}.$$

There is no monomial-position assumption in this last identity. This proves the converse in Proposition 4.1.

For the forward direction, start with a zero n and choose $b = 4^r$ arbitrarily large, exceeding all its coordinates. Then $f = (b-1)/3$ is a positive integer and can be arbitrarily large. Pack $g = \sum_i n_i \mathcal{B}^{w_i}$. Since $P(0) > 0$, at least one coordinate is nonzero, so $g \ge \mathcal{B} \ge b$. The usual finite-digit bound gives $g < b\mathcal{B}^W = \mathscr{C}$. The mask, coefficient test, and carry identity prove the required divisibility. This completes all parts of the replacement lemma.

5 The unchanged ten-integer bridge

Introduce the remaining eight supplied integer variables h, k, l, w, x, y, z, m. With $X, Y, b, \mathscr{C}$ as above, define

$$\begin{aligned}
L_b &= lY, \quad U = 2L_bX, \quad V = 4gwY, \quad W_b = bw, \\
A &= U(V+1), \quad B_b = 2X + 1, \quad C = B_b + (A-2)h, \\
P_n &= U^2V, \quad K = X + 1 + k(P_n - 2), \\
S &= 2A - 5, \quad T = 3W_bC - 2(W_b^2 - 1), \\
R_2 &= (P_n^2 - 4)K^2 + 4, \\
\Delta_b &= 8\mathscr{C}^3gK^2 - g^2\{32(C - KL_b)^2\mathscr{C}^3 + g^2K^2\}, \\
O &= f^2l^2\Delta_b, \quad \xi = zx,
\end{aligned}$$

$$
\begin{aligned}
&D = (A^2-4)C^2+4, \quad E = C^2D\xi,\\
&F = 4(A^2-4)E^2+1, \quad G^* = 2(A+2)(A-2)^2E^2-1,\\
&\Omega = 16E, \quad H^* = C+(B_b-C)F+\Omega Fy,\\
&I^* = ((G^*)^2-1)(H^*)^2+1.
\end{aligned} \tag{10}
$$

These are Sun's bridge and modified Lucas certificate [3, Propositions 6.1–6.2 and Theorems 3.1, 4.1], in the source-31 factorization. The local subscript b merely distinguishes these symbols from earlier bounds. In particular, L_b is not the coefficient bound L, and B_b is not the coding base $\mathcal{B}$.

For clarity, the exact imported contract is the following.

Proposition 5.1 (Sun bridge contract, specialized)**.** *For source polynomials satisfying Proposition 4.1 and cap (8), the formulas (10) have these properties.*

1. *A zero of P yields f, g, h, k, l, w, with f as large as needed and $A > 2$, such that b and R_2 are integer squares, $S \mid T$, $C = u_{B_b}(A)$, and $O > 0$.*
2. *Given these positive bridge witnesses, the modified Lucas interpolation theorem permits $\xi > 0$ divisible by any prescribed positive integer, and permits an arbitrary residue class for y modulo ξ, while making DFI^* a square.*
3. *Conversely, b a square, $O > 0$, $\xi \neq 0$, DFI^* and R_2 squares, and $S \mid T$, imply a nonnegative zero of P.*

Here $u_j(A)$ denotes the Lucas sequence with $u_0 = 0, u_1 = 1$, and $u_{j+1} = Au_j - u_{j-1}$. It occurs in the proof contract, not as an operation in the final equation. The reverse bridge first obtains $g > 0$, $K \neq 0$, and $0 < g < 2\mathscr{C}$ from $\Delta_b > 0$, then recovers the power-of-two and binomial conditions and invokes the source lemma. Its numerical hypotheses $3 \mid X$, $X > 3b$, and $Y > \max(b, 256)$ were re-established above, before decoding. Thus the use of the bridge does not assume that radix-five positions are still present.

On forward witnesses, Δ_b is a positive integer multiple of g, and $g \geq b \geq 4$. Hence

$$\Delta_b \geq 4, \qquad O = f^2l^2\Delta_b \geq 4.$$

Also $S = 2A-5$ is an odd, nonzero integer at *every* integer assignment, without any solution hypotheses.

6 The final polynomial and proof of equivalence

For reviewability we restate the source-31 finalizer and its proof. This section is inherited structure with the new source code substituted.

6.1 A divisibility and nonvanishing merger

Let

$$\pi(v) = (v^3-v-1)(v^2+23), \qquad \Sigma = \xi S^5, \qquad r = T+Sy,$$

and define

$$\Theta = (r^3 - rS^2 - S^3)(r^2+23S^2). \tag{11}$$

The polynomial π has no rational root and has a root modulo every positive integer. Here is a short proof of the latter fact. For an odd prime $p \neq 23$, the cubic has nonzero discriminant -23. If it has a root, that root is simple. Otherwise it is irreducible; Frobenius acts as an even three-cycle, so its

discriminant is a nonzero square modulo p, and v^2+23 has a simple root. At $p=23$, the cubic has the simple root 3. Hensel lifting handles all odd prime powers. Since $-23 \equiv 1 \pmod 8$, it is a square in the 2-adic integers. The Chinese remainder theorem completes the argument. The rational-root test rules out a rational root of the cubic, and the quadratic is positive on $\mathbb{Q}$.

Since $S \neq 0$, equation (11) never gives $\Theta = 0$. If $\Sigma \mid \Theta$, then $S \mid T$: otherwise select a prime with $\alpha = v_p(S) > \beta = v_p(T)$. The two homogeneous factors in Θ have valuations exactly 3β and 2β, by their unique least-valuation terms. Thus $v_p(\Theta) = 5\beta < 5\alpha$, contradicting $S^5 \mid \Theta$.

Conversely, if $S \mid T$ and $\xi \neq 0$, write $T = Sq$, select a root c of π modulo $|\xi|$, and prescribe $y \equiv c - q \pmod \xi$. Then $\Sigma \mid \Theta$. This is precisely the residue freedom retained by Proposition 5.1.

6.2 Merging two square requirements

Define three radicands and three weights:

$$\begin{aligned} r_0 &= b\{1+(3O-4)b^2z^2\}, & r_1 &= R_2, & r_2 &= DFI^*, \\ v_0 &= 1, & v_1 &= r_0, & v_2 &= r_0(P_n^2+K^2+3). \end{aligned} \tag{12}$$

If $z \neq 0$ and r_0 is a square, its two displayed factors are coprime. Therefore b is a signed square; $b \equiv 1 \pmod 3$ excludes zero and negative squares. The possible zero second factor would force $|b| = 1$, also giving $b = 1$. Thus b is a positive square, and $1+(3O-4)(bz)^2$ is a square. Since $bz \neq 0$, positivity follows immediately: if $O \leq 0$, this integer is at most $1-4(bz)^2 < 0$. Hence $O > 0$.

Conversely, on forward witnesses $O \geq 4$. The integer $d = 3O-4 > 0$ is $2 \pmod 3$, hence nonsquare. A positive Pell unit gives a solution $u^2 - dv^2 = 1$ with $v > 0$ divisible by b: raise the unit to its finite order modulo b. Writing $v = bz$ gives $z > 0$ and makes r_0 a square. Prescribe this z as a divisor of ξ in the interpolation theorem and set $x = \xi/z$. No extra supplied unknown is used.

6.3 Why the small weights force each square

Whenever $z \neq 0$, one has $|r_0| > 1$. If $r_0 = 0$ or $r_0 = \pm 1$, the defining factors force $|b| = 1$; then $b = 1, f = 0, O = 0$ and $r_0 = 1 - 4z^2 \leq -3$, a contradiction. Moreover

$$\sqrt{|r_1|} \leq P_n^2 + K^2 + 2.$$

Indeed $|r_1| \leq (P_n^2+4)K^2+4$, and subtracting this from $(P_n^2+K^2+2)^2$ leaves $P_n^4+K^4+P_n^2K^2+4P_n^2 \geq 0$. Consequently

$$|v_1| > \sqrt{|r_0|}, \qquad |v_2| > \sqrt{|r_0|} + |v_1|\sqrt{|r_1|}. \tag{13}$$

If a signed sum $\sum_{j=0}^{2} \varepsilon_j v_j \sqrt{r_j}$ is rational, take an automorphism of the multiquadratic splitting field and the largest index whose nonzero radical term it changes. The magnitude of that term is at least $|v_j|$, while the sum of all earlier magnitudes is strictly smaller by (13). Its change cannot cancel. Thus every nonzero radical is fixed by every automorphism and is rational; a rational square root of an integer is integral. Zero radicands are already squares. The argument uses complex absolute values and therefore does not presuppose that the radicands are nonnegative or independent.

6.4 The radical-free equation

Set

$$\begin{aligned} Z_n &= \Sigma m - \Theta, & A_j &= \Sigma^2 v_j^2 r_j \quad (0 \le j \le 2), \\ V_n &= Z_n^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z_n^2 A_2 - 4A_0 A_1. \end{aligned} \tag{14}$$

The requested polynomial is

$$\boxed{F_{10} = W_n^2 - 16 V_n^2 Z_n^2 A_2.} \tag{15}$$

Equations (1)–(15), the literal weights, and the supplied fixed integer tuple completely specify it. The formal identity

$$F_{10} = \prod_{\varepsilon \in \{-1,1\}^3} \left(Z_n + \sum_{j=0}^{2} \varepsilon_j \sqrt{A_j} \right) \tag{16}$$

is used only in its proof. The evaluator contains no radicals. An exact expansion of the right side has 35 integer polynomial terms in Z_n, A_0, A_1, A_2; the independent symbolic check compares it coefficientwise to the left side.

Proof of Theorem 1.1, apart from the degree. Start with $P(n) = 0$. The replacement source lemma and the positive bridge give the required square and divisibility conditions, with $O \ge 4$. The Pell construction gives $z > 0$ making r_0 square. The interpolation theorem supplies $\xi > 0$ divisible by z, and a y in the prescribed congruence class, such that r_2 is square and $\Sigma \mid \Theta$. Put $x = \xi/z$. All three r_j are now integer squares. Select m so that $\Sigma m - \Theta$ is the negative of one signed sum of their weighted integer roots multiplied by Σ. One factor of (16) vanishes, giving $F_{10} = 0$.

Conversely suppose $F_{10} = 0$ at an arbitrary integer assignment. We always have $S \neq 0$ and $\Theta \neq 0$. If $\Sigma = 0$, all A_j vanish and (15) becomes $\Theta^8 \neq 0$, impossible. Thus $\xi = zx \neq 0$, in particular $z \neq 0$. Divide a vanishing factor of (16) by Σ; it gives a rational weighted sum of the roots of the r_j. The separation argument makes all of them integer squares, and the same factor gives $\Sigma \mid \Theta$. The merger therefore gives $S \mid T$, b square, and $O > 0$. The reverse Sun bridge and Proposition 4.1 give a nonnegative zero of P.

Composition with (2) proves $\exists F_{10} = 0 \leftrightarrow \neg\,\mathrm{Con(ZFC)}$. All imported results and the displayed arguments are ordinary mathematics formalizable in ZFC. Classical negation gives the stated equivalence for lack of solutions. This argument neither assumes nor establishes which side is true in the standard model. □

7 Exact degree and the small evaluator

7.1 A symbolic degree calculation

Let

$$n = 36W + 8 = \deg N, \qquad c = 4W + 1 = \deg \mathscr{C}.$$

All degrees in this section give each of the ten supplied unknowns degree one and each fixed coefficient degree zero. Propagating degrees through the displayed formulas gives the following useful checkpoints:

Expression	Degree	Expression	Degree
$D_c(\mathcal{B})$	$16W$	J_c	$32W+4$
N, S_c	n	R_c	$2n$
X	$3n$	Y	$2n$
A	$7n+3$	C	$7n+4$
P_n	$12n+4$	K	$12n+5$
Δ_b	$28n+3c+14$	O	$28n+3c+18$
r_0	$28n+3c+23$	r_1	$48n+18$
r_2	$616n+340$	Σ	$35n+17$
v_2	$52n+3c+33$	A_2	$790n+6c+440$

These are attained degrees, not merely syntactic upper bounds. For example, the geometric-sum term uniquely dominates J_c; the leading term of Δ_b is supplied by $-32g^2(C-KL_b)^2\mathscr{C}^3$, with KL_b dominating C; and K^2 uniquely dominates the parenthesis in v_2. Following the nonzero leaders verifies the subsequent rows.

The scaled third radicand A_2 strictly dominates A_0, A_1, Z_n^2. In the final norm its fourth power has coefficient one and dominates all other terms. Consequently

$$\begin{aligned}\deg F_{10} &= 4(790n+6c+440)\\ &= 3160n+24c+1760\\ &= 113856W+27064\\ &= \boxed{69\,554\,088\,184}.\end{aligned}$$

This argument depends on the nonzero positive β_S, not on an accidental cancellation among the four native compiler ports.

As a separate finite attainment certificate, substitute every supplied unknown by one indeterminate and compute the degree and leading coefficient modulo three primes. At the claimed degree the leading coefficients are 95 (mod 101), 909 (mod 1009), and 668382 (mod 1000003). A single nonzero such coefficient already proves attainment once the degree upper bound is known. The checker computes these from the actual fixed tuple.

7.2 Circuit representation and operation convention

The JSON circuit is an ordered list of literal, fixed, variable, addition, subtraction, and multiplication nodes. Every arithmetic node references earlier nodes. Only the ten listed variable leaves are quantified. The 79 fixed leaves are β_S and the canonical nonliteral coefficient recipes, bound to the actual integer payloads. Fixed leaves and reuse are free; each binary addition, subtraction, and multiplication, including multiplication by a numeral, is counted. Powers are expanded into multiplication nodes. The count does not assert that arithmetic on million-bit coefficients takes unit bit complexity.

For source evaluation, put $t_0 = \mathcal{B}^W$ and $t_i = \mathcal{B}^{W-w_i}$. Then

$$D_c(\mathcal{B}) = \sum_{\alpha} c_\alpha t_0^{4-|\alpha|} \prod_{i=1}^{31} t_i^{\alpha_i}.$$

A finite shared Horner expansion evaluates this polynomial. The pure powers $\mathcal{B}^{w_i}$ and $\mathcal{B}^{W-w_i}$ are constructed with 234 multiplications. The deterministic builder starts with binary powers, merges frequently shared pairs of disjoint bit components, and reuses the resulting products. Its certificate labels each node by its exact exponent; multiplication adds labels, which proves every target power without relying on numerical tests.

The geometric sum is computed with the division-free identities

$$G_{2r} = G_r(1 + \mathcal{B}^r), \qquad G_{2r+1} = G_{2r} + \mathcal{B}^{2r}, \qquad G_r = \sum_{j=0}^{r-1} \mathcal{B}^j.$$

The resulting live arithmetic counts are

$$484 + 25 + 824 = \boxed{1333}.$$

An independent evaluator expands the entire D_c block exactly over $\mathbb{Z}[\mathcal{B}, \beta, K_0, M_C, M_F]$, obtaining the same 675 symbolic terms as the literal definition. It does not import the circuit builder. It also compares all named formulas on modular assignments with the actual coefficient tuple.

7.3 A conservative comparison variant

The archive retains `equation10_sun_finalizer.json`, which uses the same new source coding but Sun's literal four-square, sextic finalizer. Its exact degree is 1 323 756 220 880 and its evaluator has 1 383 operations. This variant separates the new additive-coding argument from the repository's later finalizer optimizations and gives an additional review route.

8 A secondary improvement: twelve positive unknowns at degree 938

This section uses the complete source-31 coefficient tuple already bundled with this report, with port aliases $\beta = \texttt{beta}, K_0 = \texttt{K}, M_C = \texttt{MC}, M_F = \texttt{MF_source}$. The last mask is shifted. It does not use the Sun coefficient or the new exponent code. The inherited normalized blank-K215 compiler interface is the one proved in source 31's `coefficients/COEFFICIENT_PROOF.md`; source 34's positive reductions are explicitly uniform in such complete tuples. The new step is a separator of formal degree nine, replacing the previous degree-23 separator. All symbols in this section are local.

The twelve supplied unknowns are

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

All other symbols below are polynomial abbreviations. Set

$$\begin{aligned}
q &= \beta J + 1, & \omega &= w + q, & C &= q - F - \alpha,\\
X &= \omega q, & Y &= sq^3, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H, & \ell &= EkY,\\
A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, & b_0 &= X + ac,\\
I &= ic^2, & D &= 1 + \Delta I^2, & Q &= (\Delta I)^2,\\
d &= Q - 1, & v &= cT, & b &= c + RD.
\end{aligned}$$

Here the abbreviation R, and the two transport expressions, are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_k &= k - hE - R, \qquad N_t = (K_0 + \omega)C + q - F - z(q-1).
\end{aligned}$$

The new majorant and translated carrier are

$$\boxed{M = k(ik^3 + T)^2}, \qquad \delta = (a+2)c - \ell,$$
$$\sigma = b + M\delta + v(a+2)I, \qquad L = HM + 1, \qquad W = b_0 + H(t + \sigma).$$

Write $e = Q(Dv^2 + b^2) - 1$ and $B_1 = 2Qvb$. The polynomial norm is

$$N = \prod_{\substack{\tau=\pm\sqrt{A},\ g=\pm\sqrt{G} \\ f=\pm\sqrt{D}}} [d(W - HM\tau - Lg)^2 - H^2 e + H^2 B_1 f]. \tag{17}$$

This is an integer polynomial; the formal sign product introduces no quantified unknowns. It is valid also for square, zero, or dependent radicands. The scalar circuit below and the delivered JSON files give a literal evaluation using only integer additions and multiplications. The two presentations are

$$\boxed{N_k N_t (1+N) - 1 = 0}, \qquad \boxed{N_k = 1, \quad N_t = 1, \quad N = 0}. \tag{18}$$

For completeness, one scalar evaluation of the norm is

$$\begin{aligned}
n &= (HM)^2 A, & d_n &= dn, & H_Q &= H^2 Q, \\
H_P &= H_Q D v^2, & H_U &= H_Q b^2, & H_e &= H_P + H_U - H^2, \\
j &= 4 H_P H_U, & \mathcal{T} &= 2d(W^2 + L^2 G), & \mathcal{K} &= [d(W^2 - L^2 G)]^2, \\
a_t &= \mathcal{T} - 2(d_n + H_e), & b_t &= (d_n - H_e)^2 - j - \mathcal{K}, & & \\
\bar{v} &= 16 d_n j - a_t(a_t \mathcal{T} + 2b_t), & u &= b_t^2 - a_t^2 \mathcal{K}, & & \\
N &= \mathcal{K}\bar{v}^2 - u(\mathcal{T}\bar{v} - u). & & & &
\end{aligned}$$

These are the sourced trace–remainder identities with the new inputs.

Theorem 8.1. *For the complete fixed coefficient tuple specified above, ZFC proves that absence of positive integer solutions to either presentation in* (18) *is equivalent to* Con(ZFC)*. The single polynomial has exactly twelve supplied unknowns, degree 938, and a 122-operation evaluator. The system has three equations, maximum residual degree 929, and 117 operations. Binary additions, subtractions, and multiplications by numerals are all charged; reuse and system comparisons are free.*

Inherited interface, stated explicitly. For the authentic tuple, $N_k = N_t = \epsilon$, $\epsilon \in \{1, -1\}$, implies before recovering any root that

$$q \geq 16, \quad Y \geq q^3, \quad E = XY > R + 2, \quad 3q < R < q^4 - 2,$$
$$k = R + \epsilon + hE > E, \quad kY < c < k(Y+1), \quad \sqrt{G} - \sqrt{A} > kY^2.$$

The recovered four normalized Pell equations, with integral roots and $g \equiv b_0 \pmod{H}$, imply a blank K215 halt; the signed-quotient extension restores $\epsilon = 1$. Conversely a halt admits canonical positive roots g, τ, f, y, γ, with $X = 2^R$. These are the inherited compiler assertions, not claims established by finite tests in the present report. The coordinate shift is complete: $2^R/q > q$, so the original w_{old} can be replaced by the positive integer $w = w_{\text{old}} - q$. This is the shift proved in source 37.

The new majorant estimate. The shifted coordinate gives $X \geq q(q+1) > q^2$, hence $R < X^2$. The bounds $c < (17/16)kY$ and $a+2 < (17/16)E$ follow at once. Crucially,

$$\frac{R(a+2)^2c^4}{k^8Y^2} < 1. \tag{19}$$

Indeed the left side equals

$$\frac{R}{X^2}\left(\frac{a+2}{E}\right)^2\left(\frac{c}{kY}\right)^4\left(\frac{E}{k}\right)^4 \leq \left[\frac{q}{q+1}(1+q^{-2}+2q^{-5})(1+q^{-3})^2\right]^2 < 1.$$

For the last inequality, set $x = q^{-2}$. The two parentheses have product at most $(1+2x)(1+x)^2 < 1+5x < 1+1/q$, since $q \geq 16$.

Let $Z = v\sqrt{D}+b$. Every complex auxiliary root satisfying $dy^2 = Q(vf-b)^2 - 1$, $f^2 = D$, obeys the sharp estimate

$$|y| < Z + v + 1. \tag{20}$$

To see this, $\Delta > R+2$, $I^2 \geq c^4$ give $Q - b = \Delta I^2(\Delta - R) - c - R > 0$, so $b \leq d$, and $\sqrt{D} < \Delta I < d$. Therefore

$$|y|^2 \leq \frac{QZ^2+1}{d} < \left[Z\left(1+\frac{1}{d}\right)\right]^2, \qquad Z/d < v+1.$$

The strict middle inequality has numerator $QZ^2 - d > 0$.

Since $D < (a+2)^2I^2$, equation (19) yields $b/(kY^2) < i^2k^7 + 1$. Also

$$\frac{v\sqrt{D}}{kY^2} < (17/16)^4 EYiTk^2 < \frac{(17/16)^4}{X}iTk^4 < iTk^4.$$

Using $v/(kY^2) < T$ and $1/(kY^2) < 1$, we obtain

$$\frac{Z+v+1}{kY^2} < i^2k^7 + iTk^4 + T + 2 < i^2k^7 + 2iTk^4 + kT^2 = M.$$

Thus $|y| < MkY^2 < M(\sqrt{G} - \sqrt{A})$. Moreover $\sqrt{A} > \ell = XkY^2$, so $M\sqrt{A} > XZ > 2Z$. The latter will ensure positivity in the converse.

Soundness. The scalar norm satisfies $N \equiv \square \pmod 4$: write $\mathcal{T} = 2x_0$ and $\mathcal{K} = x_0^2 - 4d^2W^2L^2G$ in the final formula. Thus a zero of the product in (18) forces $N = 0$ and $N_k = N_t = \epsilon \in \{1, -1\}$. A vanishing factor gives coherent algebraic roots with

$$x = (W - Lg)/H = M\tau + y, \quad \tau^2 = A, \quad g^2 = G, \quad f^2 = D, \quad dy^2 = Q(vf-b)^2 - 1.$$

Every conjugate of x has magnitude less than $M\sqrt{G}$. If g were irrational, an automorphism reversing it would produce two such x's at distance $2L\sqrt{G}/H > 2M\sqrt{G}$, impossible. Hence g is rational and integral, and x is rational. A sign change of an irrational τ would require two auxiliary roots of magnitude less than $M\sqrt{A}$ to compensate a difference $2M\sqrt{A}$. Thus τ is integral, y is rational, and $f = (e - dy^2)/B_1$ is rational and integral.

The sourced rational Pell saturation lemma now applies to

$$(\Delta I)^2(vf-b)^2 - ((\Delta I)^2 - 1)y^2 = 1$$

and makes y, hence x, integral. This lemma says that for integers $S \geq 2, V$, a rational y satisfying $S^2V^2 - (S^2-1)y^2 = 1$ is integral. Source 24 proves it by reducing the denominator to a common Pell unit and using divisibility of its cosine sequence.

Since $\delta, \sigma > 0$, $W > 0$; a negative g would contradict the bound on x. Reducing $W - Lg = Hx$ modulo H gives $g \equiv b_0 \pmod{H}$. Its quotient γ is positive because $g > (a+1)c > b_0$. The normalized compiler interface now supplies a blank K215 halt; its ordinary strong coordinate is $i_{\mathrm{ord}} = \Delta i$. No independent sign changes of dependent radicals were assumed.

Completeness. From a canonical positive compiler witness and the complete w-shift, put

$$t = \gamma + M(g - \delta) + M\tau + y - b - v(a+2)I.$$

We have $0 < \delta < ac < g$, because $\ell/c > E/2 > 2$. Also $(a+2)I < 2\sqrt{D}$ and $M\tau > 2(v\sqrt{D} + b) > b + v(a+2)I$. Therefore t is a positive integer. Its definition makes $W = Lg + H(M\tau + y)$, so one norm factor vanishes and $N_k = N_t = 1$. Composing both directions with the calibrated K215 refutation theorem, and negating existential solvability, proves the stated ZFC equivalence.

Exact degree. Scale the twelve variables by a formal parameter. The degrees of $E, H, c, I, D, Q, b, M, v$ are respectively $6, 6, 5, 11, 34, 46, 38, 9, 6$. The positive-leading roots g, τ, f have degrees $11, 11, 17$, with

$$g - \tau = \delta - k/2 + O(\lambda^{-1}), \qquad (a+2)I - f = I/(2(a+2)) + \text{lower-degree terms}.$$

The four positive-f factors in (17) have degree 119. In the negative-f factors the leading auxiliary terms cancel. Their remaining leading terms have degrees 116,116,114,107: the first three come from the main radical signs, and the last from $Hv((a+2)I - f)$, of degree 17. All competing squared terms have degree at most 104. Thus $\deg N = 4(119) + 2(116) + 114 + 107 = 929$.

More explicitly, with $q_0 = \beta J$, $k_0 = \eta + \zeta$, $\omega_0 = w + q_0$, the nonzero homogeneous leader is

$$N_{929} = -2^{46} i^{43} k_0^{103} T^5 s^{152} \omega_0^{69} q_0^{549} (q_0 - F)^8.$$

This is nonzero for every authentic $\beta > 0$. Since $\deg N_k = 7$ and $\deg N_t = 2$, the product has exact degree 938. The new product improves source 34's degree-1022, 124-operation form in both metrics; its system improves the corresponding degree-1013, 120-operation system. The existing degree-1081 evaluator at 108 operations retains its operation-count advantage. There is no variable-count or global-optimality claim.

The delivered checker compares 123 independent formal-radical products with the scalar circuit and verifies six complete univariate degree specializations, including three with the actual coefficient tuple. The generic runs also check the displayed leading coefficient. The full proof details are in `positive/POSITIVE_PROOF.md`. These finite checks support the algebra and do not replace the inherited compiler theorems or claim a new end-to-end formalization.

9 What the finite artifacts establish

The archive distinguishes mathematical proofs, exact finite certificates, and diagnostic evaluations.

Unbounded mathematical arguments. The full-simplex coefficient identity, arbitrary-input source bounds, carry reduction, and finalizer equivalence are proved above. The positive-majorant proof is included in this report and in a longer proof note. These are combined with explicitly identified compiler, Pell, Lucas, and machine theorems. Those imported arguments are not replaced by a successful Python run.

Exact finite certificates. The primary driver verifies all 25 pinned source-31 file hashes and the 17 additional archived proof dependencies, reconstructs the shifted quartic and all 402 weighted records, checks the actual fixed coefficient bounds and payloads, reconstructs the finite-field code, exhausts all 52 360 four-element multisets, proves the shared-power labels, and regenerates the delivered circuits byte for byte. It checks the three-radical and four-radical norm identities as exact sparse polynomials. The coefficient compiler reconstructs all four native payloads from the supplied finite data and compares their raw bytes. Its replay includes 69 230 basis checks and all 943 tile checks.

Independent algebra and degree checks. The separate primary evaluator proves the entire D_c source-encoding polynomial identity by sparse expansion, then compares every named formula at 60 assignments over three prime fields. Exact nonzero leading coefficients certify degree attainment. The secondary checker performs 123 formal-radical product comparisons and six complete univariate polynomial specializations, three using the actual coefficient tuple modulo the selected primes. Its symbolic leading-form proof is given in the report.

What is not certified. These scripts do not search the infinite integer domain or establish whether a zero exists. They do not rebuild the imported Lean development or mechanize the newly written unbounded proofs. Independent mathematical review is included as audit notes, but those notes are not a substitute for a proof-assistant kernel. The audits were separate AI-assisted reviews during this work, not external peer review. The theorem claimed is an ordinary ZFC theorem supported by the displayed proof and its cited dependencies.

9.1 Reproduction

From the extracted top-level directory, with Python 3.10 or later:

```
python3 reproduce.py
```

The primary checks use the Python standard library and require no network access. The optional secondary checker uses NumPy:

```
python3 positive/check_majorant.py
```

Build this PDF with a standard LaTeX installation:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The prebuilt PDF is included. The file `README.md` gives the inventory and integration notes. Manifests record the pinned source paths, Git blob hashes, SHA-256 values, and coefficient conventions. The upstream payloads and their license notices are preserved.

10 The general lesson and remaining opportunities

The old radix-five choice pays for an exponent vector in which every coordinate could independently range from zero to four. A quartic only uses vectors with *total* degree at most four. This is the structural excess removed by the additive code.

More generally, for a degree-δ polynomial with $\delta \geq 2$ in n source variables, a B_δ set of $n+1$ elements,

one translated to zero, makes

$$e(\alpha) = \sum_{i=1}^{n} \alpha_i w_i$$

injective on $|\alpha| \le \delta$. With reversal target δW and coefficients $(\delta - |\alpha|)! \prod_i \alpha_i! \, p_\alpha$, the same proof isolates $\delta! P$. Bose–Chowla supplies such sets in a cyclic group of order $q^\delta - 1$ for prime powers $q \ge n + 1$. For fixed δ, this permits exponent ranges of order n^δ, rather than the exponential positional scale $(\delta + 1)^n$.

For this particular quartic, the next concrete mathematical target would be a smaller certified 32-element B_4 set, or a support-aware code proven to separate the source support from *all* competing quartic monomials. Merely checking pairwise collisions within the support is not sufficient. Any smaller W can be substituted into the present proof, with primary degree $113856W + 27064$, provided the full certificate and source bounds are retained.

Reducing the number of supplied unrestricted integer unknowns below ten would require a different contribution. Nothing here achieves that. Likewise, the best remaining simplification may depend on whether the intended reader values low degree, a short evaluator, small coefficients, or a short formal proof. The delivered results give two concrete improvements with complete definitions, while keeping those objectives distinct.

A The literal positive quartic parent

The parent variables, in the order used for the 31 positive weights, are

$$\begin{aligned}&(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ &\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).\end{aligned}$$

All are positive integers in the parent. The symbols in this appendix are local and should not be identified with the ten final variables. Define the affine abbreviations

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C = q - F - \alpha.$$

The 22 equations, whose left side minus right side gives the residuals in order, are

$$\begin{aligned}L &= q^2 - 1,\\ N &= q(L + 1),\\ X &= wq,\\ a &= E + sN,\\ E &= X(a - E),\\ c &= k(a - E) + \eta,\\ P &= E(c - \eta) + k,\\ \tau^2 &= P(P - k) + 1,\\ D &= X + ac + \gamma H,\\ \Delta &= a^2 + H,\\ C_2 &= c^2,\\ D^2 &= \Delta C_2 + 1,\\ k &= jc - V + 1 + hE,\\ jc - V &= (q - F)(N - q) - CL + (M_C + qM_F)J,\end{aligned}$$

$$
\begin{aligned}
(K_0 + w)C + (q - F) - z(q-1) &= 1,\\
F_2 &= f^2,\\
V &= of - c,\\
g &= y^2 - V^2,\\
I &= iC_2,\\
B_{\mathrm{aux}} &= I^2,\\
B_{\mathrm{aux}} &= \Delta(F_2 - 1),\\
y^2 - B_{\mathrm{aux}} g &= 1.
\end{aligned}
$$

After substituting the affine abbreviations, each residual is quadratic. Replacing each listed variable by $n_i + 1$ and summing the residual squares gives (1). The delivered `parent_quartic31.json` is the machine-readable authority; the verifier reconstructs the weighted coefficients directly from it.

B Artifact map and attribution

Path	Role
`report.tex`, `report.pdf`	Main definitions, proofs, comparisons, and reproducibility instructions.
`data/INSTANCE.json`	Authoritative new fixed instance, parameters, degree, and paths.
`data/weights_certificate.json`	Literal weights, finite-field recipe, and exhaustive multiset certificate.
`generated/equation10.json`	Primary 1 333-operation, ten-integer polynomial.
`generated/equation10_sun_finalizer.json`	New source code with the literal Sun finalizer.
`code/build_equation.py`, `code/shared_powers.py`	Deterministic circuit construction and shared exponent evaluation.
`code/weights.py`	Independent finite additive-code construction and exhaustive verification.
`code/check_equation.py`	Source integrity, coefficient reconstruction, norm identities, and degree checks.
`code/check_coefficients.py`	Rebuilds the four complete compiler coefficients from the supplied tiles and machine.
`code/check_circuit_independent.py`	Exact sparse source identity and independent scalar transcription checks.
`positive/`	Twelve-positive-variable circuit, builder, checker, binding, and full proof.
`proofs/`	Full coefficient-coding argument and independent mathematical audit notes for both improvements.
`upstream/source31/`	Exact pinned dependencies, complete payloads, compiler data, and original license notices.
`upstream/dependencies/`	Additional inherited proof excerpts and the calibrated K215 endpoint.

This supplement was prepared with ChatGPT assistance for incorporation into ProveIt. The mathe-

matical inputs, authorship of the general theorems, and inherited source files are credited explicitly. The new explanatory text and scripts are intended for distribution under the repository's MIT No Attribution license, with all inherited third-party notices retained. See `NOTICE.md`, `LICENSE`, and the machine-data notices in the archive.

References

[1] R. C. Bose and S. Chowla, *Theorems in the additive theory of numbers*, Commentarii Mathematici Helvetici **37** (1962/63), 141–147. Original report version: https://www.cs.umd.edu/~gasarch/COURSES/858/S13/BoseChowla.pdf.

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[3] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2026 preprint. In particular Proposition 2.1, Theorems 3.1 and 4.1, Propositions 6.1–6.2, and Section 7. https://arxiv.org/html/2609.01594v2.

[4] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. Definition 1.5 and Lemma 1.6 give another explicit statement of the arbitrary-position mask property. https://arxiv.org/html/2506.20909v3.

[5] Vladimir Reshetnikov and ProveIt contributors, *Con(ZFC) Diophantine forms*, repository source collection, reviewed at commit `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`. Packages 24–31 supply the inherited compiler and Pell interfaces; packages 31–37 supply the reviewed comparison constructions. https://github.com/VladimirReshetnikov/ProveIt/tree/22e75dd6b87b1b199bb08c778d7cf075ae5aae08/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms.